FirstStep — Presentation Slide Structure

Slide plan for the 10-minute pitch, derived from *docs/technical-documentation.md*. 2026 Shenzhen FinTechathon · International Track · Topic B: Wealth Advisory Agent.

Target: ~10 min · 16 slides (~30–40s each) · Audience: FinTechathon judges. Narrative arc: Problem (1–3) → Product (4–6) → Mechanism (7–11) → Evidence (12–16).

Section A — Hook & Problem (slides 1–3)

Slide 1 — Title

- Project name **FirstStep**; tagline: “A trustworthy first advisor for first-time investors.”
- Meet **Sprout**, the guide.
- Competition line: 2026 Shenzhen FinTechathon · International Track · Topic B.
- Footer disclaimer: *Educational simulation. Not financial advice. No real money moves.*
- Team names / repository URL.

Slide 2 — The Problem

- First-time investors rarely fail by picking the wrong fund. They fail by:
- – Investing money they’ll need soon
- – Carrying expensive debt
- – Buying products riskier than they can bear
- – Selling in the first crash
- The twist: an LLM advisor makes this *worse* if it can be talked into anything — promising returns, recommending unsuitable products, placing unapproved orders.

Slide 3 — Our Thesis (the one big idea)

- **“The LLM talks; a deterministic guardian decides.”**
- The model chooses words and which tool to call. Every number, product and order comes from tested Python.
- Buying needs a one-time token that only the user’s Confirm click creates — the model never sees it.
- *This is the slide the whole deck hangs on.*

Section B — What It Does (slides 4–6)

Slide 4 — The Advisory Journey

- The closed loop, each step a logged tool call:
- 1. Money check (readiness) 2. Risk level C1–C5 3. Product comparison in plain words
- 4. A plan (range of outcomes, never one number) 5. Orders you confirm 6. Monitoring through real history
- Visual: stage flow diagram.

Slide 5 — Three Demo People

- **Amina** (21, credit-card debt): Sprout says “not yet” → savings plan.
- **Leo** (27, balanced C3): builds a plan, confirms orders, lives through March 2020.
- **Mei** (58, careful C2): asks for all tech stocks → refused in plain words.
- Visual: phone-amina-not-yet.png, desktop-leo-plan.png, desktop-mei-refusal.png.

Slide 6 — Live Demo / Screenshots

- One question per page → dashboard (Home, Plan, Funds, Activity).
- A refusal card: names the rule, says why, offers a next step.
- Confirm dialog: every trade waits for your click.
- Visual: desktop-leo-confirm.png + desktop-mei-refusal.png.

Section C — How It Works (slides 7–11)

Slide 7 — System Architecture

- Three ways in (tap, typed message, Confirm click) → **one executor**.
- Layered design: api → agent → policy → tools → engine (pure engine), enforced by import-linter.
- Visual: architecture.png.

Slide 8 — One Path for Every Action

- Every call: validate args → **guardian check** → run tool → log to hash-chained audit.
- A refusal comes back as data (rule ID + plain message), never an exception.
- Model never sees: a number it didn't get from a tool, the place_order tool, or a confirmation token.

Slide 9 — The Two Roles & Tool Tiers

- Advisor (LLM): reads, picks tools, writes words — forbidden to guarantee or predict.
- Guardian (code): re-checks every call regardless of what the model asked.
- Tool tiers: 0 Read · 1 Propose · 2 Execute (never offered to LLM) · 3 Sensitive (cooling-off).

Slide 10 — Core Algorithms (the finance)

- Readiness: 3-month fund, no debt > 8% APR.
- Risk C1–C5: tolerance vs capacity, **lower wins** (Grable & Lytton; CSRC suitability).
- Allocation: model portfolios, concentration caps (≤40%/fund).
- **Range of outcomes:** moving-block bootstrap, 5,000 paths, 10th/50th/90th percentile — never one number.
- Worked example: Leo → ~$59,500 typical (bad $47,300 / good $72,100).

Slide 11 — Security: Defence in Depth

- Layers: input guard → stage-scoped tools → guardian → **HMAC confirm token (CONF-01)** → output filter → cooling-off → hash-chained audit.
- Maps to OWASP Top 10 for LLM Apps 2025 (LLM01 Prompt Injection, LLM06 Excessive Agency).
- **Key property: no defence depends on the model behaving.**

Section D — Proof & Close (slides 12–16)

Slide 12 — Results (headline numbers)

- 25 / 25 typed requests completed.
- **30 / 30 red-team attacks handled safely** — nothing filled without a Confirm click.
- 4 / 4 audit hash chains verified.
- 223 tests passing · 95% line coverage.
- Measured on Gemma 4 26B, open-weight, free tier — judged from the audit log, not the wording.
- Visual: results.png.

Slide 13 — How We Tested

- Task completion (25), Red team (30 attacks / 9 categories), Personas (3 end-to-end), 223 automated + Playwright E2E.
- Everything judged from the audit log; CI runs on every push.

Slide 14 — Honesty & Limits

- Funds fictional; returns are *real* monthly history 2000–2025 (French Library, FRED, World Bank).
- Outcomes are ranges from history, not forecasts — never called a backtest.
- Known limits: simplified questionnaire, no real KYC, single currency, LLM can still word things badly (output filter mitigates).

Slide 15 — Why It Matters / Differentiation

- Open-weight model → a bank could self-host and keep client data in-house.
- Compliance edits a number in rules.yaml, not in code.
- Works fully offline on a scripted model — demo/tests/CI never depend on a network.

Slide 16 — Close

- Recap the thesis: *the LLM talks, the guardian decides.*
- Repo URL · live demo link · “Every trade waits for your click.”
- Thank you / Q&A.

Design notes

- Strict 10 min: merge 8→7 and 13→12. Longer slot: split slide 10 into two.
- Visual assets in repo: docs/images/architecture.png, results.png, and 11 screenshots under docs/images/screenshots/.
- Existing deck: docs/slides/presentation.html; speaker notes: docs/presentation-notes.md — keep this structure consistent with them.